1. Determinant of a 2×2 matrix

The **determinant** of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is denoted by $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ (note the change from square brackets to vertical lines) and is defined to be the number $ad - bc$. That is:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

We can use the notation det(A) or $\mid A \mid$ or Δ to denote the determinant of A.

Find the determinants of the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 4 & -1 \\ -2 & -3 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad D = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix},$$

$$E = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \qquad F = \begin{bmatrix} -1 & 0 \\ 0 & -3 \end{bmatrix}, \qquad G = \begin{bmatrix} 1 & 2 \\ -2 & -4 \end{bmatrix}.$$

Your solution

Answer

$\mid A \mid = 1 \times 4 - 2 \times 3 = -2$ $\qquad \mid B \mid = 4 \times (-3) - (-1) \times (-2) = -12 - 2 = -14$

$\mid C \mid = 0$ $\qquad \mid D \mid = 3$ $\qquad \mid E \mid = 8$ $\qquad \mid F \mid = 3$ $\qquad \mid G \mid = -4 + 4 = 0$

2. Laplace expansion along the top row

This is a technique which can be used to evaluate determinants of any order. In principle, this method can use any row or any column as its starting point. We quote one example: using the top row.

Consider $\Delta = \begin{vmatrix} 4 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{vmatrix}$.

First we introduce the idea of a **minor**. Each element in this array of numbers has an associated minor formed by removing the column and row in which the element lies and taking the determinant of the remainder. For example consider element $a_{23} = 3$. We strike out the second row and the third column:

$$\begin{array}{ccc} 4 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \qquad \text{to leave} \qquad \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} = 4 - 3 = 1.$$

For the element $a_{31} = 3$ we strike out the third row and first column:

$$\begin{array}{ccc} 4 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \qquad \text{to leave} \qquad \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 3 - 2 = 1.$$

Task What is the minor of the element $a_{22} = 2$?

Your solution

Answer
$\begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix} = 8 - 3 = 5$

Next we introduce the idea of a **cofactor**. This is a minor with a sign attached. The appropriate sign comes from the pattern of signs appropriate to a 3×3 array:

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

(i.e. positive signs on the leading diagonal and the signs 'alternate' everywhere else.)
Each element has a cofactor associated with it. The cofactor of element a_{11} is denoted by A_{11}, that of a_{23} by A_{23} and so on.

To obtain the cofactor of an element of a 3×3 matrix we simply multiply the minor of that element by the corresponding sign from the 3×3 array of signs.
Hence the cofactor corresponding to a_{23} is

$$A_{23} = -\begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} = -1$$

and the cofactor corresponding to a_{31} is $A_{31} = +\begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 1.$

Task What is the cofactor of the element a_{22}?

Your solution

Answer
The sign in the position of a_{22} in the array of signs is $+$
Hence, since the minor of this element is $+5$ the cofactor is $A_{22} = +5$.

Cofactors are important as it can be shown that the value of the determinant of a 3×3 matrix can be found from the formula

$$\Delta = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}.$$

In words "the determinant of a 3×3 matrix is obtained by multiplying each element of the first row by its corresponding cofactor and then adding the three together". (In fact this rule can be extended to apply to **any** row or **any** column and to **any** order square matrix.)

Key Point 7

Evaluating General Determinants

If A is an $n \times n$ square matrix then : $\quad \det(A) = \sum_{j=1}^{n} a_{ij}A_{ij}$

In words:

The determinant of a square matrix is obtained by multiplying each element of row i by its corresponding cofactor and then adding these products together.

In the case of $\Delta = \begin{vmatrix} 4 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 2 \end{vmatrix}$ we have $a_{11} = 4,\ a_{12} = 1,\ a_{13} = 1,$

$$A_{11} = +\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 4 - 3 = 1$$

$$A_{12} = -\begin{vmatrix} 1 & 3 \\ 3 & 2 \end{vmatrix} = -(2 - 9) = 7$$

$$A_{13} = +\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = 1 - 6 = -5$$

Hence $\quad \Delta = 4 \times 1 \ + \ 1 \times 7 \ + \ 1 \times -5 \ = \ 6.$

Alternatively, choosing to expand along the second row:

$$\begin{aligned} \Delta &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= 1\left(-\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}\right) + 2\left(\begin{vmatrix} 4 & 1 \\ 3 & 2 \end{vmatrix}\right) + 3\left(-\begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix}\right) = 6 \qquad \text{as before.} \end{aligned}$$

Task Use expansion along the first row to find $\Delta = \begin{vmatrix} 1 & -1 & 3 \\ 0 & 2 & 6 \\ -2 & 1 & 5 \end{vmatrix}$

Your solution

Answer

$a_{11} = 1, \quad a_{12} = -1, \quad a_{13} = 3$

$$\begin{aligned} A_{11} &= +\begin{vmatrix} 2 & 6 \\ 1 & 5 \end{vmatrix} = 10 - 6 = 4 \\ A_{12} &= -\begin{vmatrix} 0 & 6 \\ -2 & 5 \end{vmatrix} = -(0 + 12) = -12 \\ A_{13} &= +\begin{vmatrix} 0 & 2 \\ -2 & 1 \end{vmatrix} = 2 + 2 = 4. \end{aligned}$$

Hence $\Delta = 1 \times 4 + (-1) \times (-12) + 3 \times 4 = 4 + 12 + 12 = 28$, as before.

3. Properties of determinants

Often, especially with determinants of large order, we can simplify the evaluation rules. In this Section we quote some useful properties of determinants in general.

1. If two rows (or two columns) of a determinant are interchanged then the value of the determinant is multiplied by (-1).

   For example $\begin{vmatrix} 4 & 3 \\ 1 & 2 \end{vmatrix} = 8 - 3 = 5$ but (interchanging columns) $\begin{vmatrix} 3 & 4 \\ 2 & 1 \end{vmatrix} = 3 - 8 = -5$ and (interchanging rows) $\begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} = 3 - 8 = -5$.

2. The determinant of a matrix A and the determinant of its transpose A^T are equal.

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = 4 - 6 = -2$$

3. If two rows (or two columns) of a matrix A are equal then it has zero determinant.

   For example, the following determinant has two identical rows:

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} = 1 \times \left(\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} \right) + 2 \times \left(- \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} \right) + 3 \times \left(\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} \right)$$
$$= -3 + 2 \times (6) + 3 \times (-3) = 0.$$

4. If the elements of one row (or one column) of a determinant are multiplied by k, then the resulting determinant is k times the given determinant:

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 8 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 3 \\ 7 & 8 & 9 \end{vmatrix}.$$

   Note that if one row (or column) of a determinant is a multiple of another row (or column) then the value of the determinant is zero. (This follows from properties 3 and 4.)

   For example:

$$\begin{vmatrix} 2 & 4 & -1 \\ 4 & 2 & 1 \\ -4 & -8 & 2 \end{vmatrix} = 2 \times \begin{vmatrix} 2 & 1 \\ -8 & 2 \end{vmatrix} + 4 \times \left(- \begin{vmatrix} 4 & 1 \\ -4 & 2 \end{vmatrix} \right) - 1 \times \begin{vmatrix} 4 & 2 \\ -4 & -8 \end{vmatrix}$$

$$= 2(12) + 4(-12) - (-24) = 0$$

   This is predictable as the 3rd row is (-2) times the first row.

5. If we add (or subtract) a multiple of one row (or column) to another, the value of the determinant is unchanged.

   Given $\begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}$, add ($2 \times$ row 1) to (row 2) gives

$$\begin{vmatrix} 1 & 2 \\ 4 + 2 \times 1 & 5 + 2 \times 2 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 6 & 9 \end{vmatrix} = 9 - 12 = -3 = \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix}$$

6. The determinant of a lower triangular matrix, an upper triangular matrix or a diagonal matrix is the product of the elements on the leading diagonal.

   As an example, it is easily confirmed that each of the following determinants has the same value $1 \times 4 \times 6 = 24$.

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{vmatrix}, \quad \begin{vmatrix} 1 & 0 & 0 \\ 2 & 4 & 0 \\ 3 & 5 & 6 \end{vmatrix}, \quad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{vmatrix}$$

Task

This task is in four parts. Consider

$$\Delta = \begin{vmatrix} 1 & 4 & 8 & 2 \\ 2 & -1 & 1 & -3 \\ 0 & 2 & 4 & 2 \\ 0 & 3 & 6 & 3 \end{vmatrix}$$

(a) Use property 2 to find another matrix whose determinant is equal to Δ:

Your solution

Answer

$\Delta = \begin{vmatrix} 1 & 2 & 0 & 0 \\ 4 & -1 & 2 & 3 \\ 8 & 1 & 4 & 6 \\ 2 & -3 & 2 & 3 \end{vmatrix}$, by transposing the matrix.

(b) Now expand along the top row to express Δ as the sum of two products, each of a number and a 3×3 determinant:

Your solution

Answer

$$\Delta = 1 \times \begin{vmatrix} -1 & 2 & 3 \\ 1 & 4 & 6 \\ -3 & 2 & 3 \end{vmatrix} - 2 \times \begin{vmatrix} 4 & 2 & 3 \\ 8 & 4 & 6 \\ 2 & 2 & 3 \end{vmatrix}$$

(c) Use the statement after property 4 to show that the second of the 3×3 determinants is zero:

Your solution

Answer

In the second 3×3 determinant, row $2 = 2\times$row 1 hence the determinant has value zero.

(d) Use the statement after property 4 to evaluate the first determinant:

Your solution

Answer

In the first 3×3 determinant column $3 = \frac{3}{2} \times$ column 2. Hence this determinant is also zero. Therefore $\Delta = 0$.

Exercises

1. Use Laplace expansion along the 1st row to determine

$$\begin{vmatrix} 3 & 1 & -4 \\ 6 & 9 & -2 \\ -1 & 2 & 1 \end{vmatrix}$$

Show that the same value is obtained if you choose any other row or column for your expansion.

2. Using any of the properties of determinants to minimise the arithmetic, evaluate

(a) $\begin{vmatrix} 12 & 27 & 12 \\ 28 & 18 & 24 \\ 70 & 15 & 40 \end{vmatrix}$ (b) $\begin{vmatrix} 2 & 4 & 6 & 4 \\ 0 & 4 & 6 & 9 \\ 2 & 1 & 4 & 0 \\ 1 & 2 & 3 & 2 \end{vmatrix}$

3. Find the cofactors of x, y, z in the determinant

$$\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ x & y & z \end{vmatrix}$$

4. Prove that, no matter what the values of x, y, z, are

$$\begin{vmatrix} y+z & z+x & x+y \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = 0$$

Answers

1. $3\begin{vmatrix} 9 & -2 \\ 2 & 1 \end{vmatrix} - 1\begin{vmatrix} 6 & -2 \\ -1 & 1 \end{vmatrix} - 4\begin{vmatrix} 6 & 9 \\ -1 & 2 \end{vmatrix} = 3(9+4) - 1(6-2) - 4(12+9) = -49$

2. (a) Take out common factors in rows and columns

$$720\begin{vmatrix} 2 & 3 & 1 \\ 7 & 3 & 3 \\ 7 & 1 & 2 \end{vmatrix} = 720\begin{vmatrix} 0 & 0 & 1 \\ 1 & -6 & 3 \\ 3 & -5 & 2 \end{vmatrix}$$ using $(-2C_3 + C_1)$ then $(-3C_3 + C_2)$.

The value of the determinant (expand along top row) is then easily found as $720 \times 13 = 9360$.

(b) Zero since (row 1) is $2 \times$ (row 4).

3. Cofactors of x, y, z are $1, -2, 1$ respectively.

1. The inverse of a square matrix

We know that any non-zero number k has an inverse; for example 2 has an inverse $\frac{1}{2}$ or 2^{-1}. The inverse of the number k is usually written $\frac{1}{k}$ or, more formally, by k^{-1}. This numerical inverse has the property that

$$k \times k^{-1} = k^{-1} \times k = 1$$

We now show that an inverse of a matrix can, in certain circumstances, also be defined.

Given an $n \times n$ square matrix A, then an $n \times n$ square matrix B is said to be the **inverse matrix** of A if

$$AB = BA = I$$

where I is, as usual, the identity matrix (or unit matrix) of the appropriate size.

Example 6

Show that the inverse matrix of $A = \begin{bmatrix} -1 & 1 \\ -2 & 0 \end{bmatrix}$ is $B = \begin{bmatrix} 0 & -\frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}$

Solution

All we need do is to check that $AB = BA = I$.

$$AB = \begin{bmatrix} -1 & 1 \\ -2 & 0 \end{bmatrix} \times \frac{1}{2}\begin{bmatrix} 0 & -1 \\ 2 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 2 & -1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The reader should check that $BA = I$ also.

We make three important remarks:

- Non-square matrices do not have inverses.
- The inverse of A is usually written A^{-1}.
- Not all square matrices have inverses.

Task

Consider $A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$, and let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a possible inverse of A.

(a) Find AB and BA:

Your solution

$AB =$ $\qquad\qquad$ $BA =$

Answer

$AB = \begin{bmatrix} a & b \\ 2a & 2b \end{bmatrix}, \qquad BA = \begin{bmatrix} a+2b & 0 \\ c+2d & 0 \end{bmatrix}$

(b) Equate the elements of AB to those of $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and solve the resulting equations:

Your solution

Answer

$a = 1, \quad b = 0, \quad 2a = 0, \quad 2b = 1.$ Hence $a = 1, \quad b = 0, \quad a = 0, \quad b = \frac{1}{2}$. This is not possible!

Hence, we have a contradiction. The matrix A therefore has no inverse and is said to be a **singular matrix**. A matrix which has an inverse is said to be **non-singular**.

- If a matrix has an inverse then that inverse is unique.
  Suppose B and C are both inverses of A. Then, by definition of the inverse,

  $AB = BA = I$ **and** $AC = CA = I$

Consider the two ways of forming the product CAB

1. $CAB = C(AB) = CI = C$
2. $CAB = (CA)B = IB = B.$

Hence $B = C$ and the **inverse is unique**.

- There is no such operation as division in matrix algebra.

  We do not write $\dfrac{B}{A}$ but rather

  $A^{-1}B$ or BA^{-1},

  depending on the order required.

- Assuming that the square matrix A has an inverse A^{-1} then the solution of

  the system of equations $AX = B$ is found by pre-multiplying both sides by A^{-1}.

$$\begin{array}{lrcl} & AX & = & B \\ \text{pre-multiplying by } A^{-1}: & A^{-1}(AX) & = & A^{-1}B, \\ \text{using associativity:} & A^{-1}A)X & = & A^{-1}B \\ \text{using } A^{-1}A = I: & IX & = & A^{-1}B, \\ \text{using property of } I: & X & = & A^{-1}B \end{array}$$

which is the solution we seek.

2. The inverse of a 2×2 matrix

In this subsection we show how the inverse of a 2×2 matrix can be obtained (if it exists).

Task

Form the matrix products AB and BA where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Your solution

$AB =$ $\qquad\qquad\qquad\qquad$ $BA =$

Answer

$$AB = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = (ad - bc)I$$

$$BA = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)I$$

You will see that had we chosen $C = \dfrac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ instead of B then both products AC and CA will be equal to I. This requires $ad - bc \neq 0$. Hence this matrix C is the inverse of A. However, note, that if $ad - bc = 0$ then A has **no inverse**. (Note that for the matrix $A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}$, which occurred in the last task, $ad - bc = 1 \times 0 - 0 \times 2 = 0$ confirming, as we found, that A has no inverse.)

Key Point 8

The Inverse of a 2×2 Matrix

If $ad - bc \neq 0$ then the 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ has a (unique) inverse given by

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Note that $ad - bc = |A|$, the determinant of the matrix A.

In words: To find the inverse of a 2×2 matrix A we interchange the diagonal elements, change the sign of the other two elements, and then divide by the determinant of A.

Task

Which of the following matrices has an inverse?

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 \\ -2 & 2 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Your solution

Answer

$|A| = 1 \times 3 - 0 \times 2 = 3; \quad |B| = 1 + 1 = 2; \quad |C| = 2 - 2 = 0; \quad |D| = 1 - 0 = 1.$

Therefore, $A,\ B$ and D each has an inverse. C does not because it has a zero determinant.

Task

Find the inverses of the matrices $A,\ B$ and D in the previous Task.

Use Key Point 8:

Your solution

$A^{-1} =$ $\qquad$ $B^{-1} =$ $\qquad$ $C^{-1} =$

Answer

$A^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix}, \ B^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \ D^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = D$

It can be shown that the matrix $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ represents an **anti-clockwise** rotation through an angle θ in an xy-plane about the origin. The matrix B represents a rotation **clockwise** through an angle θ. It is given therefore by

$$B = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Task

Form the products AB and BA for these 'rotation matrices'. Confirm that B is the inverse matrix of A.

Your solution

$AB =$

$BA =$

Answer

$$\begin{aligned} AB &= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\theta + \sin^2\theta & -\cos\theta\sin\theta + \sin\theta\cos\theta \\ -\sin\theta\cos\theta + \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \end{aligned}$$

Similarly, $BA = I$

Effectively: a rotation through an angle θ followed by a rotation through angle $-\theta$ is equivalent to zero rotation.

3. The inverse of a 3×3 matrix - Gauss elimination method

It is true, in general, that if the determinant of a matrix is zero then that matrix has no inverse. If the determinant is non-zero then the matrix has a (unique) inverse. In this Section and the next we look at two ways of finding the inverse of a 3×3 matrix; larger matrices can be inverted by the same methods - the process is more tedious and takes longer. The 2×2 case could be handled similarly but as we have seen we have a simple formula to use.

The method we now describe for finding the inverse of a matrix has many similarities to a technique used to obtain solutions of simultaneous equations. This method involves operating on the rows of a matrix in order to reduce it to a unit matrix.
The row operations we shall use are

(i) interchanging two rows
(ii) multiplying a row by a constant factor
(iii) adding a multiple of one row to another.

Note that in (ii) and (iii) the multiple could be negative or fractional, or both.

The Gauss elimination method is outlined in the following Key Point:

Key Point 9

Matrix Inverse – Gauss Elimination Method

We use the result, quoted without proof, that:

if a sequence of row operations applied to a square matrix A reduces it to the identity matrix I of the same size then the **same** sequence of operations applied to I reduces it to A^{-1}.

Three points to note:

- If it is impossible to reduce A to I then A^{-1} does not exist. This will become evident by the appearance of a row of zeros.
- There is no unique procedure for reducing A to I and it is experience which leads to selection of the optimum route.
- It is more efficient to do the two reductions, A to I and I to A^{-1}, simultaneously.

Suppose we wish to find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 2 & 7 & 7 \end{bmatrix}$$

We first place A and I adjacent to each other.

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 2 & 7 & 7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Phase 1

We now proceed by changing the columns of A *left to right* to reduce A to the form $\begin{bmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{bmatrix}$
where $*$ can be any number. This form is called **upper triangular**.
First we subtract row 1 from row 2 and twice row 1 from row 3. 'Row' refers to both matrices.

$$\begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 2 & 7 & 7 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{matrix} R2-R1 \\ R3-2R1 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

Now we subtract row 2 from row 3

$$\begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad R3-R2 \quad \Rightarrow \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

Phase 2

This consists of continuing the row operations to reduce the elements above the leading diagonal to zero.
We proceed *right to left*. We subtract 3 times row 3 from row 1 (the elements in row 2 column 3 is already zero.)

$$\begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad R1-3R3 \quad \Rightarrow \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 4 & 3 & -3 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

Finally we subtract 3 times row 2 from row 1.

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 4 & 3 & -3 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} \quad R1-3R2 \quad \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \begin{bmatrix} 7 & 0 & -3 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$$

Then we have $A^{-1} = \begin{bmatrix} 7 & 0 & -3 \\ -1 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}$
(This can be verified by showing that $AA^{-1} = I$ or $A^{-1}A = I$.)

Consider $A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 3 & -1 \\ -1 & 2 & 1 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

Use the Gauss elimination method to obtain A^{-1}.

First interchange rows 1 and 2, then carry out the operation (row 3) + $\frac{1}{2}$(row 1):

Your solution

Answer

$$\begin{bmatrix} 0 & 1 & 1 \\ 2 & 3 & -1 \\ -1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R1 \leftrightarrow R2 \quad \Rightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad R3 + \tfrac{1}{2}R1 \quad \Rightarrow \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ 0 & \frac{7}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}$$

Now carry out the operation (row 3) $-$ $\frac{7}{2}$(row 2) followed by (row 1) $-$ $\frac{1}{3}$(row 3) and (row 2) + $\frac{1}{3}$(row 3):

Your solution

Answer

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ 0 & \frac{7}{2} & \frac{1}{2} \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix} \quad R3 - \tfrac{7}{2}R2 \quad \Rightarrow \quad \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & -3 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix} \quad \begin{matrix} R1 - \frac{1}{3}R3 \\ R2 + \frac{1}{3}R3 \end{matrix} \quad \Rightarrow \quad \begin{bmatrix} 2 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} +\frac{7}{6} & +\frac{5}{6} & -\frac{1}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix}$$

Next, subtract 3 times row 2 from row 1, then, divide row 1 by 2 and row 3 by (-3).
Finally identify A^{-1}:

Your solution

Answer

$$\begin{bmatrix} 2 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad \begin{bmatrix} \frac{7}{6} & \frac{5}{6} & -\frac{1}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix} \quad R1 - 3R2 \quad \Rightarrow \quad \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} \frac{10}{6} & \frac{2}{6} & -\frac{4}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad \begin{bmatrix} \frac{10}{6} & \frac{2}{6} & -\frac{4}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ -\frac{7}{2} & \frac{1}{2} & 1 \end{bmatrix} \quad \begin{matrix} R1 \div 2 \\ \\ R3 \div (-3) \end{matrix} \quad \Rightarrow \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & -\frac{2}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{7}{6} & -\frac{1}{6} & -\frac{1}{3} \end{bmatrix}$$

Hence $A^{-1} = \begin{bmatrix} \frac{5}{6} & \frac{1}{6} & -\frac{2}{3} \\ -\frac{1}{6} & \frac{1}{6} & \frac{1}{3} \\ \frac{7}{6} & -\frac{1}{6} & -\frac{1}{3} \end{bmatrix} = \dfrac{1}{6}\begin{bmatrix} 5 & 1 & -4 \\ -1 & 1 & 2 \\ 7 & -1 & -2 \end{bmatrix}$

4. The inverse of a 3×3 matrix - determinant method

This method which employs determinants, is of importance from a theoretical perspective. The numerical computations involved are too heavy for matrices of higher order than 3×3 and in such cases the Gauss elimination approach is prefered.

To obtain A^{-1} using the determinant approach the steps in the following keypoint are followed:

Key Point 10

Matrix Inverse – the Determinant Method

Given a square matrix A:

- Find $|A|$. If $|A| = 0$ then A^{-1} does not exist. If $|A| \neq 0$ we can proceed to find the inverse matrix, as follows.
- Replace each element of A by its cofactor (see Section 7.3).
- Transpose the result to form the **adjoint matrix**, denoted by adj(A)
- Then calculate $A^{-1} = \dfrac{1}{|A|}$ adj(A).

Task

Find the inverse of $A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & 3 & -1 \\ -1 & 2 & 1 \end{bmatrix}$. This will require five stages.

(a) First find $|A|$:

Your solution

Answer

$|A| = 0 \times 5 + 1 \times (-1) + 1 \times 7 = 6$

(b) Now replace each element of A by its minor:

Your solution

Answer

$$\begin{bmatrix} \begin{vmatrix} 3 & -1 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ -1 & 2 \end{vmatrix} \\ \\ \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix} \\ \\ \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ 2 & 3 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 5 & 1 & 7 \\ -1 & 1 & 1 \\ -4 & -2 & -2 \end{bmatrix}$$

(c) Now attach the signs from the array

$$\begin{matrix} + & - & + \\ - & + & - \\ + & - & + \end{matrix}$$

(so that where a $+$ sign is met no action is taken and where a $-$ sign is met the sign is changed) to obtain the matrix of cofactors:

Your solution

Answer

$$\begin{bmatrix} 5 & -1 & 7 \\ 1 & 1 & -1 \\ -4 & 2 & -2 \end{bmatrix}$$

(d) Then transpose the result to obtain the adjoint matrix:

Your solution

Answer

Transposing, adj$(A) = \begin{bmatrix} 5 & 1 & -4 \\ -1 & 1 & 2 \\ 7 & -1 & -2 \end{bmatrix}$

(e) Finally obtain A^{-1}:

Your solution

Answer

$A^{-1} = \dfrac{1}{\text{det}(A)}\text{adj}(A) = \dfrac{1}{6}\begin{bmatrix} 5 & 1 & -4 \\ -1 & 1 & 2 \\ 7 & -1 & -2 \end{bmatrix}$ as before using Gauss elimination.

Exercises

1. Find the inverses of the following matrices

   (a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

2. Use the determinant method and also the Gauss elimination method to find the inverse of the following matrices

   (a) $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 4 & 1 & 2 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Answers

1. (a) $-\dfrac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix}$ (c) $\dfrac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$

2. (a) $A^{-1} = -\dfrac{1}{2}\begin{bmatrix} 0 & -2 & 1 \\ -2 & 4 & 2 \\ 0 & 0 & -1 \end{bmatrix}^T = -\dfrac{1}{2}\begin{bmatrix} 0 & -2 & 0 \\ -2 & 4 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

   (b) $B^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$